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CircuitVerse Project ▾ Circuit ▾ Tools ▾ Help ▾ Untitled

Main x +

CIRCUIT ELEMENTS

Search..

Input

1

0

5f

0

0

Output ▾

Gates ▾

Annotation ▾

Misc ▾

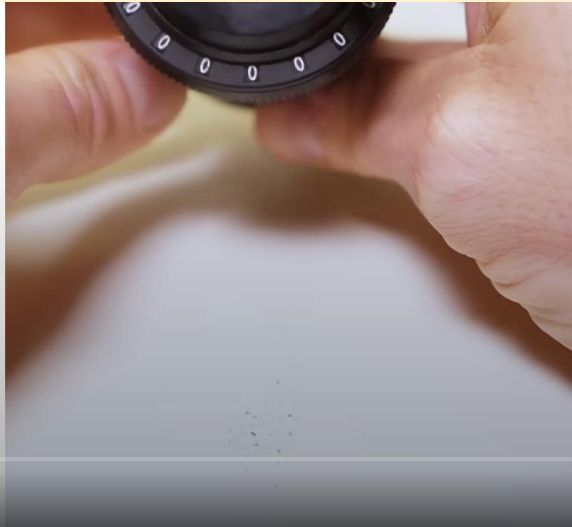
TIMING DIAGRAM

1 cycle = 1000 Units

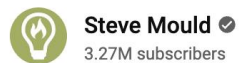
<https://circuitverse.org/simulator>



<https://www.youtube.com/watch?v=dKN6XatdVkY>



The Curta Mechanical Calculator



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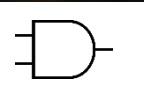
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Thanks

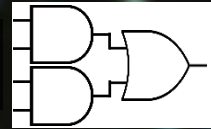
Clip



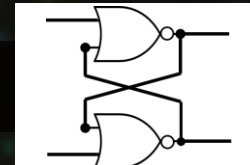
Number Systems | $(12)_{10} \rightarrow (1100)_2$
Logic Gates |



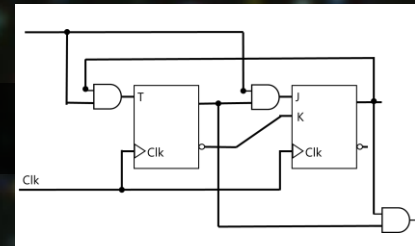
Combinational Logic |



Flip-Flop |



Sequential Logic |



RECAP

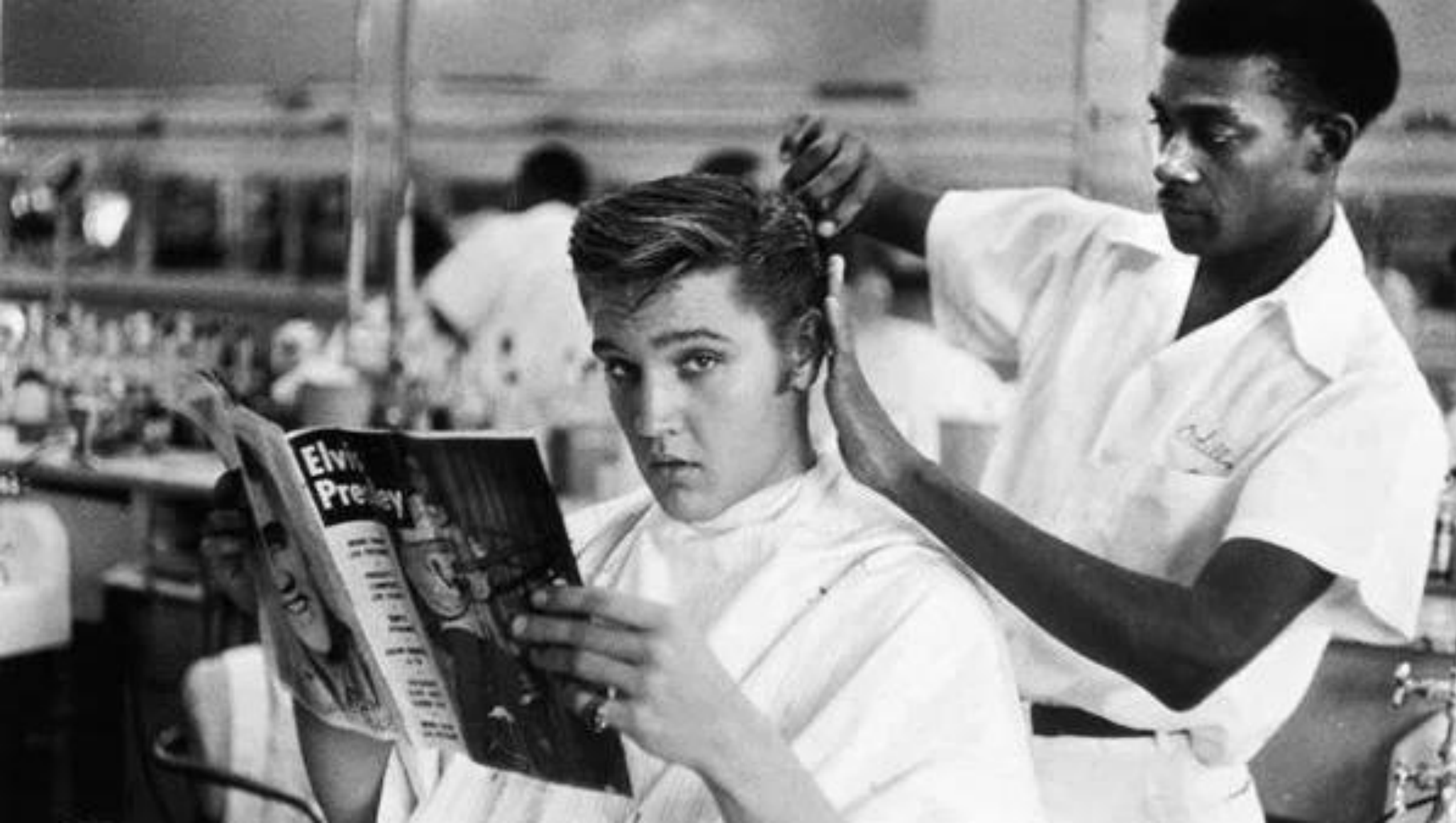
Any Boolean Function F:

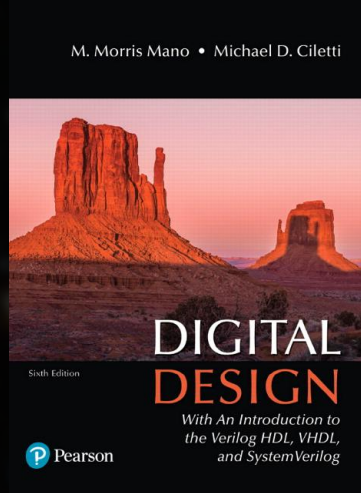
- Sum (OR) of Products (ANDs)
- Sum of **minterms** for Entries with **1**
 - ANDs-OR
 - NAND via $(F')'$
- Product (AND) of Sums (ORs)
- Product of **MAXTERMS (NOT minterms)** for Entries with **0**
 - ORs-AND
 - NOR via $(F')'$

PRACTICE: XOR

Any Boolean Function F

- Sum (OR) of Products (ANDs)
- Sum of **minterms** for Entries with **1**
 - ANDs-OR
 - NAND via $(F')'$
- Product (AND) of Sums (ORs)
- Product of **MAXTERMS (NOT minterms)** for Entries with **0**
 - ORs-AND
 - NOR via $(F')'$





Chapter 2: Boolean Algebra and Logic Gates

Chapter 3: Gate-Level Minimization

MINIMIZATION

aka. Simplification

Same Effective Design but More Efficient

MINIMIZATION

Number of Gates

Number of Inputs (2-input vs 4-input)

Number of Interconnections

Propagation Time

Cost of Gates

Circuit Area

...

A circuit may not satisfy all due to conflicting constraints!

MINIMIZATION

aka. Simplification

SoP (ANDs-OR) → Simplify → NAND
PoS (ORs-AND) → Simplify → NOR

MINIMIZATION

- I) Boolean Algebra (algebraically)
 - II) Map (Karnaugh map, K-map)
-

MINIMIZATION

- I) Boolean Algebra (algebraically)
 - II) Map (Karnaugh map, K-map)
-

ALGEBRA

A set of elements

A set of operators

A set of axioms | postulates | assumptions | definitions

ALGEBRA

A set of elements: e.g.,

$\{0,1\}$

$\{-1,0,1\}$

$\{1,2,3,4,\dots\}$

$\{\dots, -2,-1,0,1,2,\dots\}$

ALGEBRA

A set of operators

$\{+, \times\}$

$\{+, -, \times, \div\}$

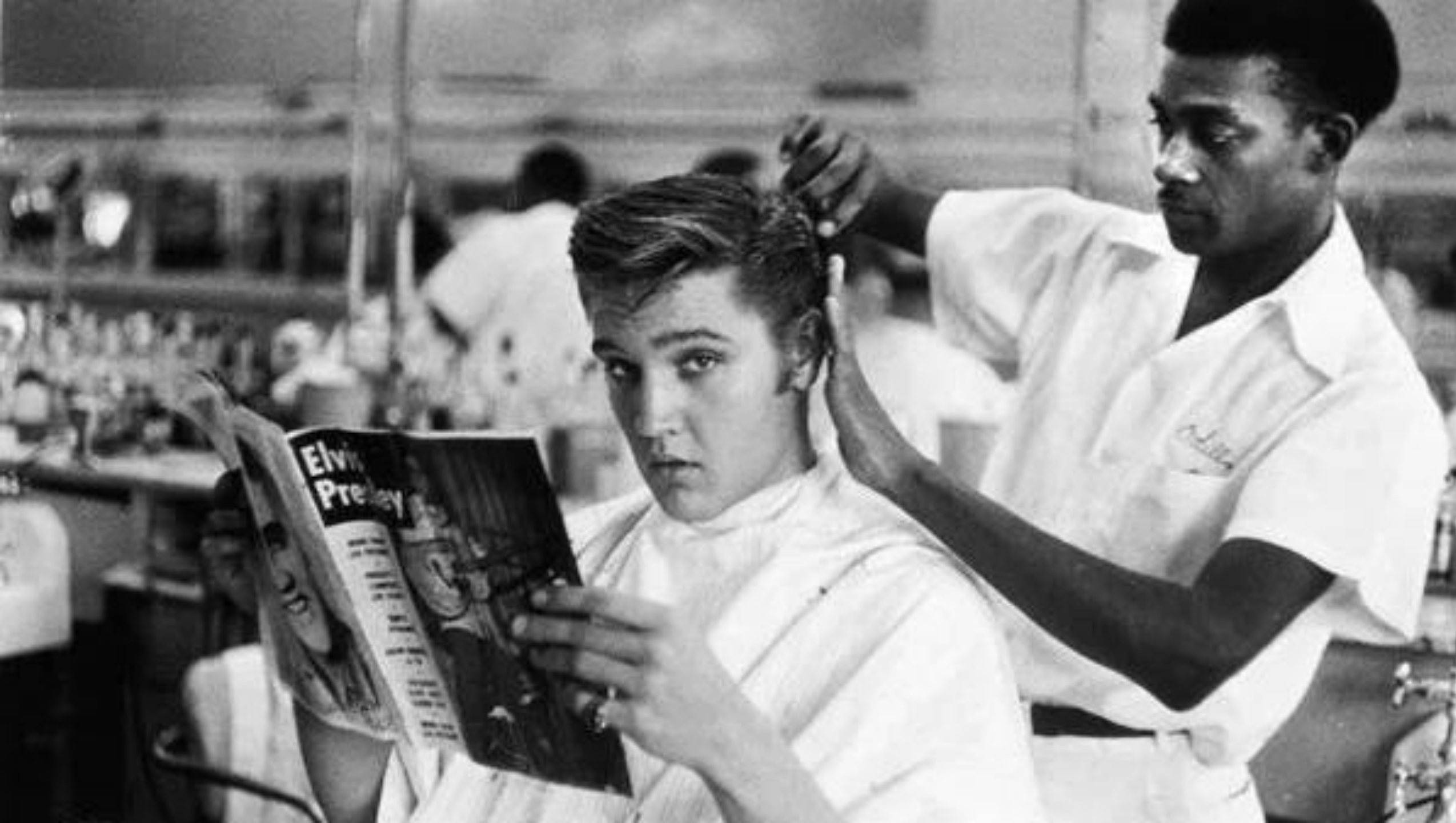
$\{\sim, \leq, \geq\}$

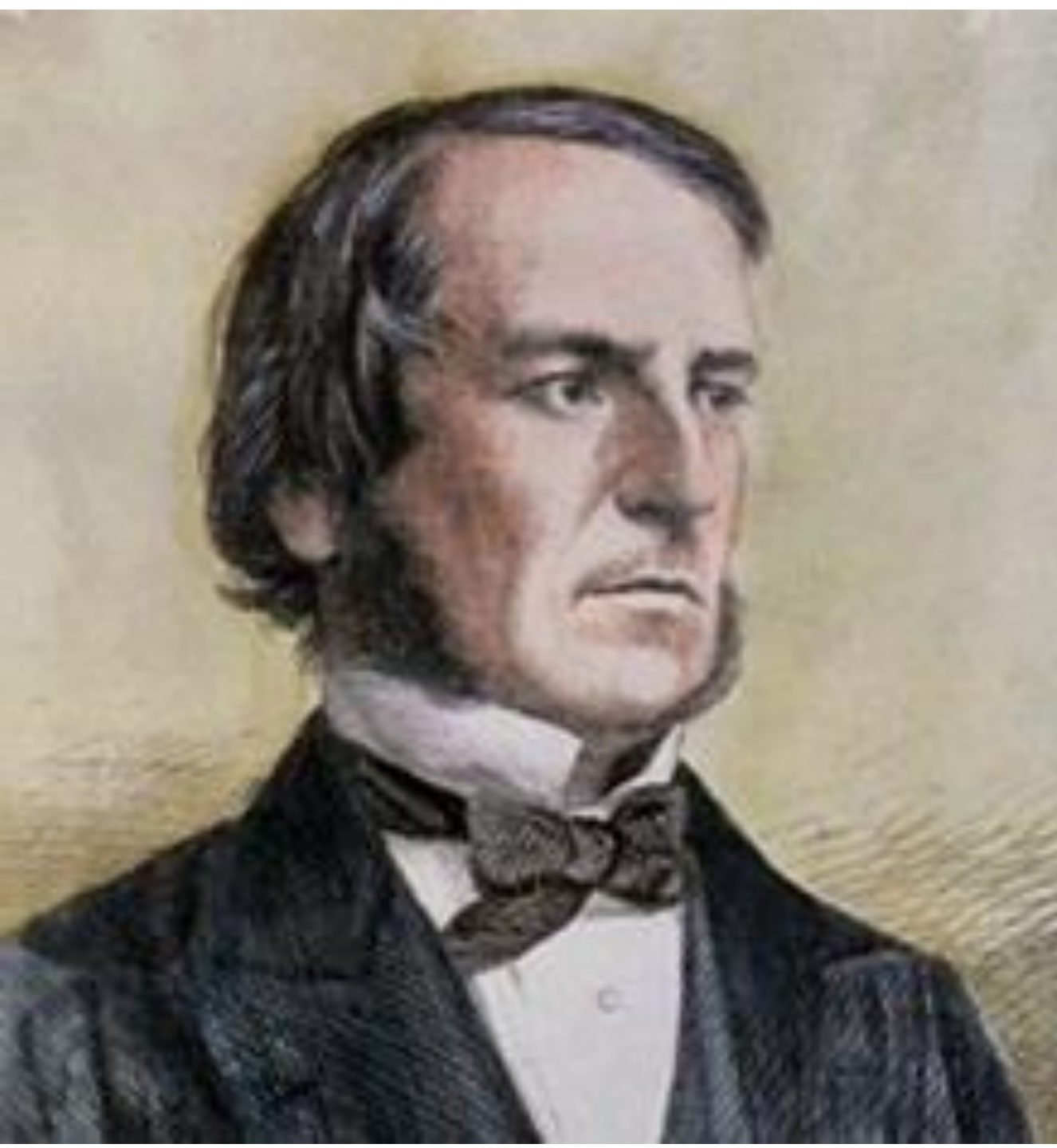
ALGEBRA

Unary: $!x$, $-x$, x' , ...

Binary: $x+y$, $x\div y$, x^y , ...

Ternary: $x?y:z$ (Elvis), x BETWEEN y AND z (SQL)





George Boole (/bu:l/)

Mathematician

Philosopher

Logician

The Laws of Thought (1854)

Boolean Algebra!



Edward Vermilye Huntington
1874 – 1952

In 1904, he put Boolean algebra
on a sound axiomatic foundation

POSTULATE

aka. Axiom, Assumption

<https://en.wikipedia.org/wiki/Axiom>

A statement that is **taken** to be **TRUE**

Serve as a premise or starting point for further reasoning

Section Sign --> Elvis, Just a Random Operator



I. CLOSURE

A set is closed with respect to an operator $\S$ iff the result of $\S \in S$:

Unary: $x \in S: \S x \in S$

Binary: $x, y \in S: x \S y \in S$

if and only if



I. CLOSURE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$\S = +$ (addition), $-$ (subtraction), $\times$ (multiplication), $-$ (negation), $!$ (fact)

S is closed with respect to $+, -, \times, -, !$

I. CLOSURE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\S = \wedge \text{ (power)}$$

?

I. CLOSURE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = ^\wedge (\text{power})$$

S is NOT closed with respect to $^\wedge$ for $2^{(-1)} \notin S$

I. CLOSURE

$$S = \{0,1\}$$

$$\S = + \text{ (OR)}, \times \text{ (AND)}, ' \text{ (NOT)}$$

S is closed with respect to $+, \times, '$

I. CLOSURE

$$S = \{0,1\}$$

$$\xi = - \text{(negation)}$$

S is NOT closed with respect to $-$ for $(-1) \notin S$

II. COMMUTATIVE

A binary operator $\S$ on a set S is commutative iff for all $x, y \in S$:

$$x \S y = y \S x$$

($x \S y$ may or may not be in S)

II. COMMUTATIVE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\S = + \text{ (addition), } \times \text{ (multiplication)}$$

+ and $\times$ are commutative on S for $x+y=y+x$ and $x\times y=y\times x$

II. COMMUTATIVE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = - \text{ (subtraction)}$$

- is NOT commutative on S for $x-y \neq y-x$

II. COMMUTATIVE

$$S = \{0,1\}$$

$$\S = + \text{ (OR)}, \times \text{ (AND)}, \uparrow \text{ (NAND)}, \downarrow \text{ (NOR)}, \oplus \text{ (XOR)}, \odot \text{ (XNOR)}$$

All are commutative on S for $x \S y = y \S x$

III. ASSOCIATIVE

A binary operator $\S$ on a set S is associative iff for all $x, y, z \in S$:

$$x \S (y \S z) = (x \S y) \S z = x \S y \S z$$

Intuitively, when you don't need parenthesis to indicate the order!

III. ASSOCIATIVE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$\xi = +$ (addition), $\times$ (multiplication), $-$ (subtraction)

$+$, $-$, $\times$ are associative on S for

$$x \pm (y \pm z) = (x \pm y) \pm z = x \pm y \pm z$$

$$x \times (y \times z) = (x \times y) \times z = x \times y \times z$$

III. ASSOCIATIVE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = ^{\wedge} (\text{power})$$

?

III. ASSOCIATIVE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\S = ^\wedge \text{ (power)}$$

$^\wedge$ is NOT associative on S for $2^\wedge(1^\wedge 3) \neq (2^\wedge 1)^\wedge 3$

III. ASSOCIATIVE

$$S = \{0,1\}$$

$$\xi = + \text{ (OR)}, \times \text{ (AND)}$$

$+, \times$ are associative on S for

$$x + (y + z) = (x + y) + z = x + y + z$$

$$x \times (y \times z) = (x \times y) \times z = x \times y \times z$$

Dagger Sign --> Just Another Random Operator



IV. DISTRIBUTIVE

If $\S$ and $\dagger$ are two binary operators on a set S , $\S$ is distributive over $\dagger$ iff:

Left Distributivity: $x \S (y \dagger z) = (x \S y) \dagger (x \S z)$

Right Distributivity: $(y \dagger z) \S x = (y \S x) \dagger (z \S x)$

IV. DISTRIBUTIVE

If $\S$ and $+$ are two binary operators on a set S , $\S$ is distributive over $+$ iff:

$$\text{If } \S \text{ Commutative: } x \S (y + z) = (x \S y) + (x \S z) = (y + z) \S x$$

IV. DISTRIBUTIVE

$$S = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

+ (Addition), $\times$ (Multiplication)

$$x + (y \times z) \stackrel{?}{\iff} (x + y) \times (x + z) \stackrel{?}{\iff} (y \times z) + x$$

$$x \times (y + z) \stackrel{?}{\iff} (x \times y) + (x \times z) \stackrel{?}{\iff} (y + z) \times x$$

IV. DISTRIBUTIVE

$$S = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

+ (Addition), $\times$ (Multiplication)

$$x + (y \times z) \neq (x + y) \times (x + z) \neq (y \times z) + x$$

$$x \times (y + z) = (x \times y) + (x \times z) = (y + z) \times x$$

IV. DISTRIBUTIVE

$$S = \{0, 1\}$$

+ (OR), $\times$ (AND)

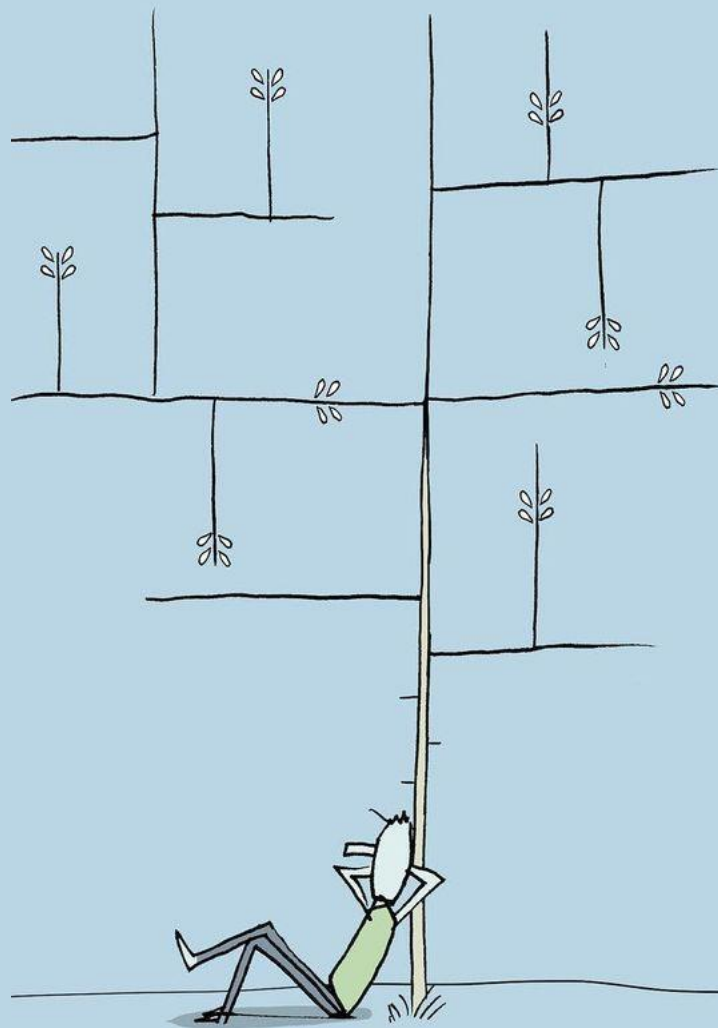
$$x + (y \times z) \stackrel{?}{\iff} (x + y) \times (x + z) \stackrel{?}{\iff} (y \times z) + x$$

$$x \times (y + z) \stackrel{?}{\iff} (x \times y) + (x \times z) \stackrel{?}{\iff} (y + z) \times x$$



<https://www.yuvalrob.com/>

yuvalrob



V. IDENTITY

$e \in S$ is an identity element w.r.t binary operator $\S$ iff for all $x \in S$:

$$x \S e = x = e \S x$$

V. IDENTITY

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = + \text{ (addition), } \times \text{ (multiplication)}$$

$$e_+ = 0 : x + 0 = 0 + x = x$$

$$e_\times = 1 : x \times 1 = 1 \times x = x$$

V. IDENTITY

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\S = - \text{(subtraction)}$$

$$e = ?$$

V. IDENTITY

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$\S = -$ (subtraction)

$$e = 0$$

$$x - 0 \neq 0 - x \neq x$$

NO identity element for $-$ (subtraction) in S

V. IDENTITY

$$S = \{0,1\}$$

$$\xi = + \text{ (OR)}, \times \text{ (AND)}$$

$$e_+ = 0 : x+0=0+x=x$$

$$e_x = 1 : x \times 1 = 1 \times x = x$$

VI. INVERSE



humans were originally created with four arms, four legs and a head with two faces. Fearing their power, Zeus split them into two separate parts, condemning them to spend their lives in search of their other halves.

VI. INVERSE

For all $x \in S$, there should be $y \in S$ w.r.t binary operator $\S$ iff:

$$x \S y = e_{\S} = y \S x$$

We denote $y = x^{-1}$ and $x = y^{-1}$

VI. INVERSE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = + \text{ (addition)}$$

$$x + (-x) = (-x) + x = e_+ = 0$$

$$x^{-1} = -x$$

VI. INVERSE

$$S = \{..., -2, -1, 0, 1, 2, ...\}$$

$$\xi = \times \text{ (multiplication)}$$

S does not have the inverse property for $\times$ since

$$2 \times \frac{1}{2} = \frac{1}{2} \times 2 = e_x = 1 \text{ but } \frac{1}{2} \notin S$$

VI. INVERSE

$$S = \{0,1\}$$

$$\S = + \text{ (OR)}, \times \text{ (AND)}$$

?

VI. INVERSE

$$S = \{0,1\}$$

$$\S = + \text{ (OR)}, \times \text{ (AND)}$$

$$1 + ? = ? + 1 \neq e_+ = 0$$

$$0 \times ? = ? \times 0 \neq e_\times = 1$$

A painting of a woman with a needle in her chest. The woman is depicted from the chest up, with her head tilted slightly to the right. She has a pale, almost white complexion and her eyes are closed. A thin needle is inserted into her upper chest, with a small amount of red blood visible at the entry point. Her right hand is raised, with fingers spread, as if she is holding or feeling the needle. The background is a deep blue with intricate, golden-yellow spiral patterns that resemble stylized waves or clouds. The overall style is somewhat abstract and expressive, with visible brushstrokes and a somber, contemplative mood.

Healing. -- Zara Mandana Fard
<https://www.baraneh.com/#/when-i-lost-hope/>